

Dr Daksh (daksh-daksh.github.io)

Postdoctoral Researcher, IMS SAS, Bratislava, Slovakia

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Postdoctoral Researcher & Research Computer Scientist at **IMS SAS, Bratislava**, specializing in **machine learning, deep learning, computer vision, and AI** for scientific and biomedical applications. Experience in supervised, unsupervised, self-supervised, and reinforcement learning, with a focus on **multidimensional image analysis and correlative microscopy using deep learning**. Research interests include **multimodal and generative AI, diffusion, and foundation models, and developing automated scalable learning frameworks** for complex scientific data.

Research Experience

Postdoctoral Researcher, Institute of Measurement Science (IMS), Department of Imaging Methods, Slovak Academy of Sciences (SAS), Bratislava, Slovakia 01/09/2026 - Present

- Working on biomedical MRI image datasets using deep learning.
- Developing an automated toolkit for optimization, segmentation, and classification of multi-label objects.

PhD Researcher & affiliated with CRC 1066, MPIP Mainz 15/10/2021 - 31/03/2026 **PhD Awarded on: 26/02/2026**

- Worked on automated image registration for correlative microscopy using deep learning.
- Developed a Python tool for training-free multimodal image registration of light and electron microscopy, along with a supervised image preprocessing tool for generating precise image pairs.
- Applied deep learning and machine learning techniques to image and microscopy data analysis, including the generation of super-resolution microscopy images.

Web Group Coordinator, Max Planck PhDnet, MPG 01/01/2023 – 01/01/2025

- Developed and maintained the PhDnet website and mailing lists.

Software Developer, VMDC, SH, India 01/08/2020 – 30/09/2021

- Designed, developed, and maintained automated pipelines for ultrasound image classification, preprocessing, and diagnostic analysis.
- Applied deep learning and computer vision techniques to medical image analysis, developing and curating diverse image datasets for model training, validation, and evaluation.
- Developed data preprocessing and augmentation workflows to improve dataset quality and support robust model development.
- Contributed to the development and evaluation of machine learning models for automated analysis and classification of ultrasound images.

Research Fellow, CSIE & IEO Department, NTNU, Taiwan 01/05/2018 – 31/07/2020

Jointly worked on a project with NVIDIA AI Tech, Taiwan.

- Worked on optical defect detection using supervised deep learning for digital holography.
- Developed data augmentation approaches using GANs, including AC-GAN and DC-GAN, for deep learning based on digital holograms.
- Applied deep learning for image analysis and generation of diverse holographic image datasets.

Scientific Co-worker, University of Greifswald, Germany 01/06/2017 – 28/02/2018

- Worked on computational modeling and simulations of biomolecules.

Software Developer, VMDC, SH, India 10/05/2016 – 30/06/2017

- Designed and maintained webpages, mailing lists, and network infrastructure supporting organizational and research activities.
- Designed and developed a clinical toolkit for patient record management and automated prediction of expected delivery dates using ultrasound images from pregnant patients.

Setup Configuration Specialist, Aon Hewitt Pvt. Ltd., India 01/11/2012 – 31/07/2014

- Selected among 5,000 students prior to the completion of the final semester.
- Worked with IBM Mainframes, Lotus Notes, C++, ASP.NET (C#), VB, and SQL.

EDUCATION

Max-Planck Institute for Polymer Research (MPIP) & Johannes Gutenberg University Mainz (JGU), Mainz, Germany

PhD - Computer Science (Interdisciplinary) 15/10/2021 – 31/03/2026

Advisor: Prof. Dr. Katharina Landfester & Dr. Ingo Lieberwirth

National Forensic Sciences University (AIR – 159 in Computer Sciences), Gandhinagar, India

MS - Forensic Nanotechnology (Nano-engineering) 14/07/2014 – 30/05/2016

Graduated in First class with Distinction

Lovely Professional University, Phagwara, India

B.Tech (Honors) - Computer Science & Engineering 15/07/2008 – 30/05/2012

Graduated in First class with Distinction

Patent

Multimodal Training-Free Registration Using Mutual Image Information.

Application submitted for patent & commercialization with Max Planck Innovations 2025.

Conference Talks

Multi-resolution Cross-modality Image Registration Using Unsupervised Deep Learning Approach. D Daksh, A Kaltbeitzel, K Landfester, I Lieberwirth. Microscopy and Microanalysis: Microscopy Society of America 2023.

Unsupervised Deep Learning approach for image registration in Correlative Microscopy for the localization of Nanoparticles. D Daksh, A Kaltbeitzel, G Glaßer, I Lieberwirth, K Landfester. Invited Speaker, BIO Web of Conferences, European Microscopy Congress 2024.

Correlative Microscopy Strategies for the Identification of Intracellular Nanoparticles and their Cellular Processing. I Lieberwirth, S Han, A Kaltbeitzel, G Glaßer, D Daksh, K Landfester. Microscopy and Microanalysis: Microscopy Society of America 2024.

Publications

TFUDL-CLEM: A Training-Free Unsupervised Deep Learning Registration Method for Correlative Light and Electron Microscopy. D. Daksh, A. Kaltbeitzel, G. Glaßer, K. Landfester, I. Lieberwirth. iScience, Cell Press (Under peer review).

A Weakly Supervised Pre-processing Pipeline for Multi-Modal Image Pair Generation for Image Registration. D. Daksh, A. Kaltbeitzel, G. Glaßer, K. Landfester, I. Lieberwirth. The Visual Computer, Springer Nature (Under peer review).

Symposium Presentations & Talks

Microscopy data analysis: machine learning and the BioImage Archive. EMBL Heidelberg, Germany May 2022

First EMBL Imaging centre Symposium: Enabling imaging across scales. EMBL Heidelberg, Germany May 2022

Oral short talk

SFB 1066 and 1278 Joint Summer school. Fulda, Germany July 2022
Poster Presentation

Joint Symposium of RMaP and SFB 1066. Mainz, Germany July 2022
Poster Presentation

EMBO workshop: from Cryo-EM to multi-scale modelling. EMBL Heidelberg, Germany February 2023
Poster Presentation

Deep Learning & computer vision (DLCV) school. Genova, Italy June 2023
Selected among 1% with full scholarship to attend. Won the first prize for the most novel idea and best presentation.

Joint Symposium of SFB1066 and TRR319. Mainz, Germany April 2024
Poster Presentation

Internal Symposium of SFB 1066. Mainz, Germany July 2024
Poster Presentation

International Symposium of SFB 1066. Mainz, Germany January 2025
Poster Presentation

Certificates

Optimizing writing strategies. MPIP Mainz, Germany February 2024

German B1. In-house course in MPIP Mainz, Germany September 2024

Python for HPC. MPCDF Garching, Germany November 2024

Parallel Computing With Matlab. MPCDF Garching, Germany December 2024

Workshop on AI and Research in MPG. Berlin, Germany December 2024

Skills

Programming Languages & Tools: Python, PyTorch, Pandas, Cuda, Anaconda navigator with different IDEs, Matlab, Napari, ImageJ, Bash, Git, LaTeX, Keras, Automated ML/DL pipelines, TensorFlow, Scikit-learn, Scipy, Numpy, C, C++, C#, .Net, SQL, Database Managemnet system, HTML, PHP, IBM mainframes, Lotus notes, LLMs.

Operating systems: Windows, Linux, MacOS.

Languages: English (Fluent), Hindi (Native), German (B1/B2), French (Basic).